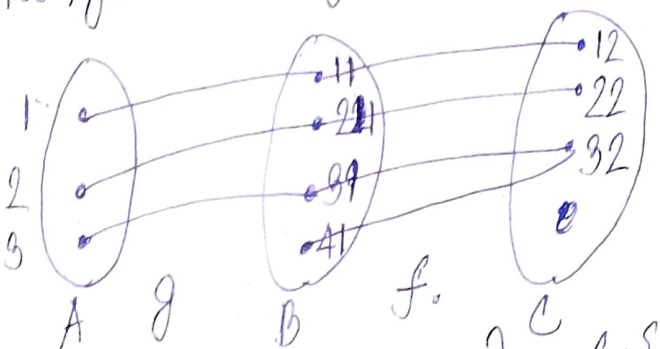
37) If f and fog are onto, does it follow that g is onto? Justify your answer.

Solution: Let $f: B \to C$ and $g: A \to B$. By Defn. 10, $fog: A \to C$

By Defn. 7, $\forall z \in C, \exists y \in B$ s.t. $f(y) = z$.
By Defn. 7, $\forall z \in C, \exists x \in A$, s.t. $f(g(x)) = z$.
Suppose, g is not onto. By Defn. 7, $\exists p \in B$ s.t. $\forall x \in A, g(x) \neq p$.



$g: \{1,2,3\} \to \{11,21,31,41\}$ $f: \{11,21,31,41\} \to \{12,22,32\}$
$fog: \{1,2,3\} \to \{12,22,32\}$

$A: \{1,2,3\}$ $B: \{11,21,31,41\}$ $C: \{12,22,32\}$

$f(11) = 12$ $f(21) = 22$ $f(31) = f(41) = 32$

It's easy to see that f is onto.

$g(1) = 11$ $g(2) = 21$ $g(3) = 31$.

$f(g(1)) = f(11) = 12$ $f(g(2)) = f(21) = 22$ $f(g(3)) = f(31) = 32$

It's also easy to see that fog is onto.

But, g is not onto, since the element 41 in B has no mapping in A.

38) Find fog and gof, where $f(x) = x^2+1$ and $g(x) = x+2$, are functions from R to R.

Solution: $fog = f(g(x)) = f(x+2) = (x+2)^2+1 = x^2+4x+5$

$gof = g(f(x)) = g(x^2+1) = (x^2+1)+2 = x^2+3$

39) Find $f+g$ and fg for the functions given in Exercise 38.

Solution: $f+g = (x^2+1)+(x+2) = x^2+x+3$

$fg = (x^2+1)(x+2) = x^3+2x^2+x+2$